Question 1

Planning (15 marks)

Defining the problem (3 marks)

P1 *p* is the independent variable or vary *p.* [1]

P2 *A* is the dependent variable or determine *A* for different *p.* [1]

P3 Keep the incident amplitude constant (allow volume of sound or power/intensity output). [1]

Methods of data collection (5 marks)

M1 Labelled diagram of a workable arrangement including source of sound – glass window – detector of sound. [1]
Allowed sources: loudspeaker, bell, buzzer, siren but not musical instruments.
Allowed detectors: microphone, sound meter, sound detector.

M2 Method of measuring pressure. Use bourdon gauge/manometer or pressure gauge [1]
Do not allow barometer or pressure meter.

M3 Method of reducing pressure. Use (vacuum) pump to withdraw air from glass window. [1]

M4 Method of measuring amplitude. Measure amplitude from oscilloscope. [1]

M5 Perform experiment in quiet room. [1]

Method of analysis (2 marks)

A1 Appropriate quantities plotted i.e. A^2 against p; or lg A against lg p or A against $\sqrt{p}$. [1]

A2 Relationship to be correct i.e. graph is a straight line through the origin. [1]
Gradient should equal 0.5 if lg A against lg p plotted.
An explicit statement is required.

Safety considerations (1 mark)

S1 Relevant safety precaution related to either the use of glass or intensity of sound, [1]
e.g. wear ear defenders, switch on sound source for short period of time.
Protection (goggles/safety screen) in case glass breaks.
Awareness of low pressures in glass.
Do not allow gloves because glass is sharp.

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Additional detail (4 marks).

D1/2/3/4 Relevant points might include: [4]
Method of ensuring that output from speaker is constant.
Method of reducing sound reflections from e.g. foam/speaker & microphone close to glass.
Window perpendicular to sound source.
Detail explaining use of oscilloscope to measure amplitude.
Difficulty in measuring amplitude at small pressures/use a loud incident source.
Control (or monitoring) of one additional variable e.g. temperature, frequency, distances.
Allow time for the temperature/pressure between the glass to stabilise.
Discussion of attenuation in air/frame/glass.

[Total: 15]

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